



Application of Generative design for a Formula Student Car Upright with Integrated Gearbox Casing for Additive Manufacture

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Summary

In this report, a detailed investigation is carried out on the prospect of designing and manufacturing a generatively designed and additively manufactured suspension upright (see Figure 2) for UoNRT's 2024 car. The objective of this project was to design a lighter single piece upright than the current assembly of multiple parts that was previously featured on the car, whilst retaining an appropriate safety factor. This reduces the amount of unsprung mass, which improves vehicle dynamics by reducing suspension reaction time.

Two functional generative designs were successfully created in both AlSi10Mg and Ti6Al4V. Maximum loading conditions were calculated, and the designs were both validated under these loads using finite element analysis. Both were found to satisfy the safety criteria established in the aims and objectives by having a factor of safety greater than two. Although the established success criteria for mass was a 20% reduction compared to the previous design, both the titanium and aluminium designs outperformed this by being 69.1% and 69.9% lighter respectively. Finally, AlSi10Mg was evaluated to be the more appropriate material for the use case, as the design was 0.8% lighter than Ti6Al4V, had a 70% lower maximum displacement, the material is 257% cheaper, and it will be easier to manufacture. Quotations were obtained from two third-party manufacturers to produce the part, and an engineering drawing detailing all of the dimensions necessary for post-machining was created.

Although material testing was carried out with the intention of producing material properties data, which would have been used in the generative design software, the data collected varied significantly from the expected values. This was likely due to porosity or inclusions within the test samples, which could clearly be seen in the fracture sites. As such, the results were disregarded and more accurate data from a third-party study were used instead for design generation.

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Nomenclature

Symbol	Quantity	Unit
m	Mass	kg
g	g-force	-
H	Height	m
W	Width	m
F	Force	N
T	Torque	Nm

Subscripts

c	Cornering
COG	Centre of gravity
t	Track (width)
v	Vertical
l	Lateral
i	Inner
o	Outer
b	Braking
r	Rotor
p	Peak
m	motor
a	axial

Acronyms

Acronym	Meaning
AM	Additive Manufacturing
AWD	All-Wheel Drive
CAD	Computer Aided Design
CNC	Computer Numerical Control
EDM	Electrical Discharge Machining
EV	Electric Vehicle
FEA	Finite Element Analysis
FOS	Factor of Safety
FS	Formula Student
ICE	Internal Combustion Engine
OBR	Oxford Brookes Racing
TUM	TU Munich (formula student team)
UoNRT	University of Nottingham Race Team

1. Introduction

1.1. Project Background

Formula Student (FS) is an engineering competition held in many countries worldwide every year [1]. For this competition, University teams from all over the world must construct an open wheeled, single seater formula car (see Figure 1) to a strict set of rules and regulations. The design must significantly innovate year after year, or severe penalties are applied in the design category.



Figure 1: University of Nottingham Race Team (UoNRT) 2021 Formula Student car [2]

The upright assembly (see Figure 2), which this project focuses on, is an essential component of the suspension system and is particularly difficult to design for an all-wheel drive (AWD) configured car as the mass and packaging complexity increases greatly. The upright is responsible for holding the wheel hub and bearings, as well as facilitating the transfer of power from the motor, and featuring connection points for the wishbones which attach it to the chassis. In this particular project, the upright will also be responsible for housing the gearbox. Due to the high speed, rapid acceleration, and high cornering force of these vehicles, the upright is subjected to relatively high g-forces, as well as any imperfections in the terrain creating unexpected impacts.

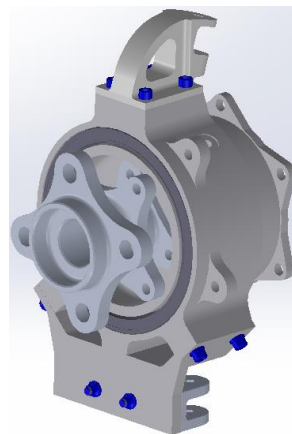


Figure 2: UoNRT 2023 Formula Student upright assembly [3]

It is advantageous to reduce vehicle mass as much as possible so that acceleration is improved, and less mass must be transferred across the car during cornering events. Furthermore, the upright and gearbox are unsprung components (see Figure 3), meaning that reduction of mass will have a significant positive impact on vehicle dynamics. This is because the inertial mass of the suspension assembly is reduced, allowing for a faster suspension reaction time. However, the challenge that usually presents itself whilst trying to optimise components for mass is that stiffness is reduced.

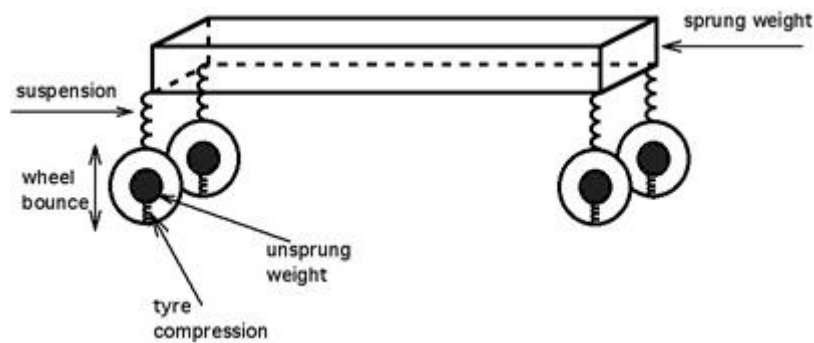


Figure 3: Diagram illustrating the difference between sprung and unsprung mass [4]

This project explores the possibility of applying generative design (see Figure 4) to create an integrated upright and gearbox component (see Figure 10). Generative design is a relatively new design methodology that leverages machine learning, artificial intelligence, and advanced algorithms to iteratively explore and evaluate thousands of design possibilities, while considering various constraints and goals, to generate an optimised design solution.



Figure 4: Example of a generatively designed and metallic additive manufactured part [5]

This generatively designed component would then be produced by metal additive manufacturing. Metal additive manufacturing is a process that enables the production of complex and high-performance metal parts by adding material layer by layer. The process offers many advantages over traditional manufacturing methods, including increased design flexibility, improved material utilization, and less material waste.

It was hoped that this generatively designed single-piece component would be significantly lighter whilst retaining appropriate stiffness, compared to the current solution of machined

billet sections attached together using bolted connections. This component was intended to be additively manufactured out of either titanium Ti6Al-4V or aluminium AlSi10Mg to reduce mass as much as possible, whilst retaining an appropriate stiffness.

1.2. Aim and Objectives

The overall aim of this work is to develop a functional generative design that integrates the upright and gearbox into one cohesive additively manufactured component for the University of Nottingham 2024 Formula Student car.

This design should be evaluated with maximum loading conditions for all expected scenarios. These loads should be applied in the worst-case combination, and the design is required to have a factor of safety (FOS) equal to or greater than two, under this loading.

The success factor for this new design was a 20% reduction compared to the previous design (see Figure 2), whilst achieving an FOS greater than or equal to two.

Material properties used within the generative design program should be validated by creating and testing 3D printed samples.

Objectives:

- Perform several generative design iterations using the generative design package in Autodesk Fusion 360 using different materials.
- Compare the differently optimised designs and the current 2023 upright to ensure that the new design is at least 20% lighter and has a FOS of at least two.
- Validate the material properties assumed by Fusion 360, by evaluating 3d printed tensile test samples.
- Create a manufacturing plan for the best design alternative.

2. Literature Review

The reduction of overall mass, specifically unsprung mass, is imperative in the design of a racing car [6]. As such, topological design, specifically generative design, has recently become a staple in motorsport industries in order to reduce mass and increase stiffness as much as possible by optimising material distribution in a given space [7]. Through the advent of metallic additive manufacturing (AM) technology, generative topological designs are able to be created as they are free from the restrictions of subtractive machining technology.

2.1. Design Technique

Generative design is a cutting-edge approach that uses algorithms and computational power to generate thousands of design options that meet a set of specified constraints and goals. It involves the use of software and machine learning algorithms to create complex designs that can be optimized for different criteria such as performance, efficiency, and cost. It has been widely used in various engineering fields such as automotive, aerospace, and architecture [8]. It is considered revolutionary in the design industry as it allows time-efficient exploration of new design possibilities that were previously very difficult and time consuming to achieve using traditional design methods [9]. Some of the benefits of generative design include faster design iteration, reduced material waste, and improved material performance per unit mass. It also enables designers to create parts that are optimized for specific manufacturing processes, such as metal AM or 3-axis computer numerical control (CNC), resulting in more efficient and cost-effective production [10].

2.2. Additive Manufacturing Methods

Metal additive manufacturing is a process of building three-dimensional objects layer by layer using metal powder, wire or other medium. The main techniques for metal additive manufacturing in industry are Powder Bed Fusion (PBF), Directed Energy Deposition (DED), Binder Jetting, and Sheet Lamination.

For PBF, a thin layer of metal powder is spread across a build plate and a laser is used to selectively melt the powder, layer by layer, until the object is formed (see Figure 5). Examples of powder bed fusion techniques include Selective Laser Melting (SLM) and Electron Beam Melting (EBM) [11].

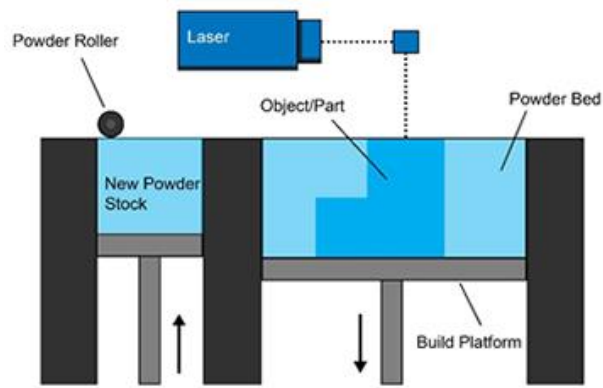


Figure 5: Powder bed fusion process illustration [12]

DED involves melting a metal wire or powder and depositing it directly onto a substrate using a nozzle or laser (see Figure 6). This process allows for the creation of large objects and can be used to repair damaged parts. Examples of directed energy deposition techniques include Laser Metal Deposition (LMD) and Electron Beam Welding (EBW) [13].

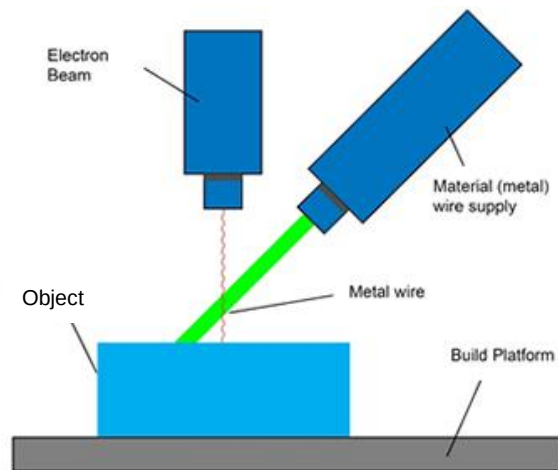


Figure 6: Directed energy deposition process illustration [12]

In the Binder Jetting technique, a layer of metal powder is spread over a build platform, and a binder material is selectively jetted onto the powder to bind it together in the desired shape (see Figure 7). After the object is printed, it is sintered in a furnace to fuse the metal particles together [14].

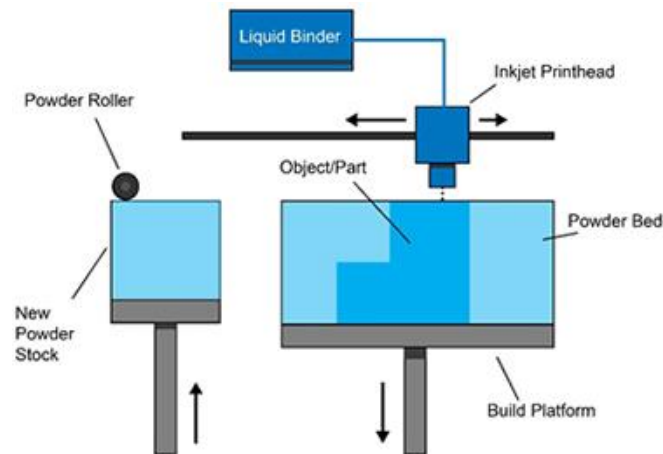


Figure 7: Binder jetting process illustration [12]

Sheet Lamination involves stacking and bonding layers of metal sheets, which have been cut and shaped using a CNC machine (see Figure 8). This process is relatively slow and less precise compared to other techniques, but it allows for relatively easy creation of large objects [15].

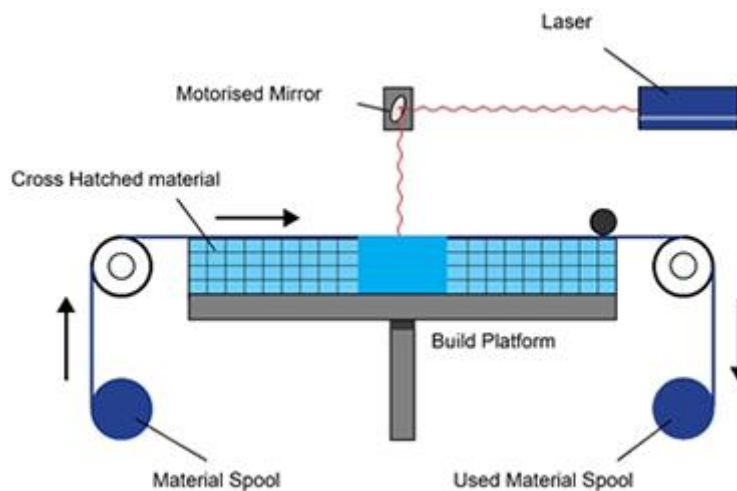


Figure 8: Sheet lamination process illustration [12]

2.3. Heat Treatment Techniques

A variety of methods have been proposed for the heat treatment of AlSi10Mg. AM machine manufacturer EOS have extensively investigated a proprietary T6 annealing process, where the part is first solution annealed at 530°C, followed by artificial aging at 165°C [16]. In the horizontal build orientation, this produced a 3.7% yield strength decrease, as well as a 0.8% increase in elongation at failure. Interestingly, in the vertical build orientation, the yield strength was increased by 8%, and elongation was increased by 4.7%. Conversely, Fiocchi has investigated microstructural changes after lower temperature annealing [17]. Samples annealed at and above 294°C were discovered to have a ruptured silicon network within the part, significantly decreasing yield strength and increasing elongation. Alternatively, samples annealed at 263°C were discovered to have a precipitated, but intact, silicon and magnesium

network. It was thought that this preservation of the silicon and magnesium network would yield a small decrease in yield strength, while increasing the ductility of the alloy significantly.

The heat treatment and subsequent material properties of Ti6Al4V are well investigated by Jaber, where it was revealed that high temperature annealing at 850°C, followed by furnace cooling, produced the best balance of 13% elongation and 932 MPa yield strength [18]. Further cooling methods were also investigated, with water quenching producing high yield strength but low ductility, and air cooling providing a result between the two.

2.4. Case Studies

Many previous projects have been carried out on the subject of additively manufactured uprights, specifically in formula student. However, these studies are focused primarily on a traditional internal combustion engine (ICE) configuration with driveshafts. TU Munich (TUM) [19] have an electric vehicle (EV) configuration but integrate the upright in a differently to how this project proposes. TUM approaches the upright in a fairly standard manner, but their unique gearbox design means that it requires only bracing from the rear (see Figure 9). Despite the size of their wheels creating an issue for wishbone attachment points, they tackled this by elongating the structure towards the chassis until there is appropriate space to attach. In a different approach, Oxford Brookes Racing (OBR) have been able to successfully implement a gearbox integrated upright solution (see Figure 10) to reduce mass by 50% [20] when compared to their previous design. It is likely that a gearbox integrated design is more optimal for mass saving and improved assembly stiffness as the main load-bearing component is braced directly.



Figure 9: TUM additively manufactured upright [19]

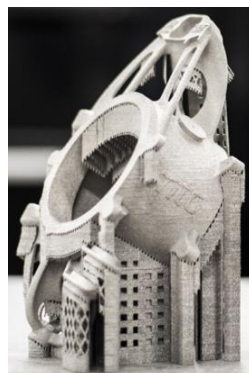


Figure 10: OBR additively manufactured upright with integrated gearbox [20]

Although it is possible to use a manual iterative process in SolidWorks where material is manually subtracted and then passed through a finite element analysis (FEA) process, as seen in [21], it can be argued that this more traditional approach to weight reduction has been antiquated by the advent of topological design software. This is because the best possible geometric solution is produced, as opposed to an educated guess [7]. Although the method proved effective and reduced the component mass by 48.61%, it is likely that this significant reduction was due to starting with a very poorly optimised design. Even if it is desirable to avoid additive manufacturing and instead opt for more traditional CNC machining, other studies have demonstrated how constraints can be effectively introduced into the generative design process in order to produce a desired outcome [22]. In this specific case, it was to reduce support material for additive manufacturing; however, a similar constraint can be applied to ensure manufacturability on CNC equipment.

This can be seen in the study carried out by Stavridis, where topological design software is used to optimise a FS upright for 3-axis CNC milling due to manufacturing resource constraints [23]. This successfully reduced component mass by 20.4% compared to an already fairly optimised component proving that, even within the constraints of traditional machining, significant mass reduction can be achieved with topology optimisation.

However, when introducing the use of additive manufacturing, even further reductions in mass can be achieved. As seen in the study carried out at the University of Ostrava, an additively manufactured upright (see Figure 11) made from the same aluminium alloy as a previously heavily optimised CNC machined upright (see Figure 12) [24]. This yielded a mass reduction of 40% with a maximum stress 53% lower than the billet machined upright, with only a small compromise of 0.09mm greater maximum deflection. From this, it can be inferred that the generative topological design in conjunction with metallic additive manufacturing will produce the lightest possible part for a given safety factor.



Figure 11: Additively manufactured generatively designed upright from the University of Ostrava [24]

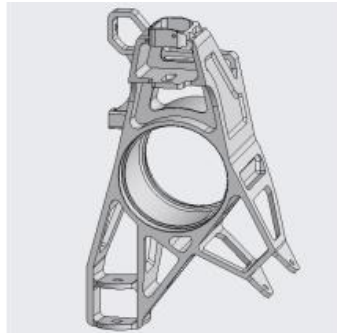


Figure 12: Previous season's CNC machined upright from the University of Ostrava [24]

2.5. Loading and Boundary Conditions

Loading conditions can be applied as reaction forces at the wishbones to ensure appropriate bracing in the design [6]. Lateral cornering forces, peak gearbox torque, bump force, deceleration force, peak motor torque, inertial motor force, and peak breaking torque must be considered and applied simultaneously to simulate an extreme loading scenario. Moments from the contact patch centroid of the tyre must be calculated and applied to the suspension pickup points as vector forces. Through experimentation, loading with reaction forces from the wishbones and fixing the bearing faces (see Figure 17) was determined to be the best loading approach during design generation. It allowed for both of the double-shear wishbone mounts (see Figure 13) to be stressed appropriately. This is as opposed to one side of the wishbone mount bearing most of the load, which is especially apparent when applying a bump load.

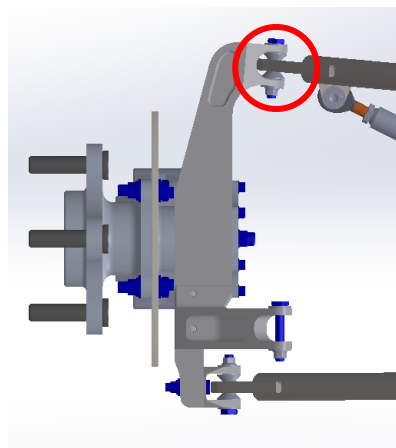


Figure 13: Example of a double shear suspension mount (circled) and wishbones [3]

3. Methodology

3.1. Material Properties Validation

In order to generate plausible designs, the material properties utilised by the program must first be validated by comparing to real world test data. As such, nine AISi10Mg bars were produced using SLM technology within the institution. Samples 6 to 9 were then annealed at 263 degrees Celsius for 10 minutes in order to further investigate the method proposed by Fiocchi. This method intends to increase elongation without drastically reducing yield strength, unlike traditional T6 high temperature annealing followed by artificial aging. Furthermore, the low-temperature annealing method requires less time and less specialised equipment so, if it yielded promising results, the manufacturability of the part would improve. All the samples were then machined to the International Organization for Standardization (ISO) tensile testing standard specified (see Appendix 4). Each sample was then tested under tension until failure, which allowed for the yield strength and elastic modulus to be determined. If these results were deemed accurate, they would then be averaged. The material properties within Fusion 360 would then be modified to produce designs that were tangibly manufacturable using this specific method and material.

3.2. Generation Geometry

The generation geometry was first created in SolidWorks. This is because the UoNRT computer aided design (CAD) models are natively in this software package, as well as the fact that there are appropriate tools to enable easier manipulation of the assembly. For the generative stage, this assembly was then exported to Fusion 360.

It was discovered early in the design process that some measure would need to be taken to attach both of the wishbone mounting bosses to each other. This is because, the program tended to detach the furthest boss, causing generation errors and program failure. Thus, separate wishbone mounts were generated (See Figure 14) and inserted into the assembly. Technically, these are not optimised for being part of the final design, but this is not a concern as the program is able to bypass these pre-generated mounts in the second pass through the software.

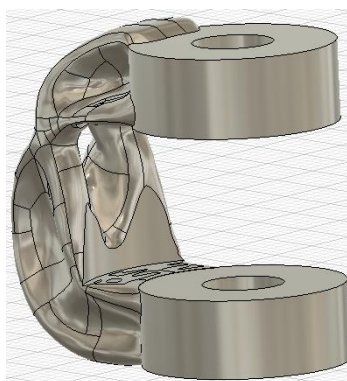


Figure 14: Separately generated double shear wishbone mount (lower mount pictured)

In Autodesk Fusion 360, the first generation was carried out by defining areas of preserved geometry (green) and obstacle geometry (red) (see Figure 15). As such, clearances for fasteners, wishbones, the brake caliper, and gearbox internals are accounted for. The preserved geometry includes a minimally thick gearbox housing (1.5mm thick) which has been verified by independent FEA, separately generated wishbone mounts, and pre-defined bosses for mounting the brake caliper and motor.

Fixed geometry and loading conditions were applied to the preserved geometry (see sections 3.3 and 3.4)

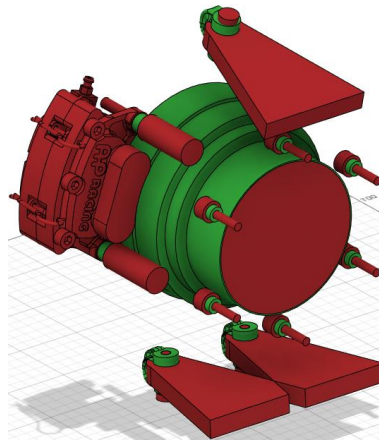


Figure 15: Fusion 360 starting geometry for producing generative upright

It was soon realised that on the first generation pass, Fusion 360 was not able to produce the best possible outcome. This was extrapolated from the fact that the FOS was above 2 in both material cases, clearly indicating that there was more material volume, and consequently mass, to be lost in the design. It was then decided that a second pass through the software would be necessary, utilising the previously generated design as a 'starting shape' (yellow) (see Figure 16) for the next generation. This allowed the program to focus more on refinement, as it already had a reasonably optimised shape to begin with, rather than having to create a base design from scratch.

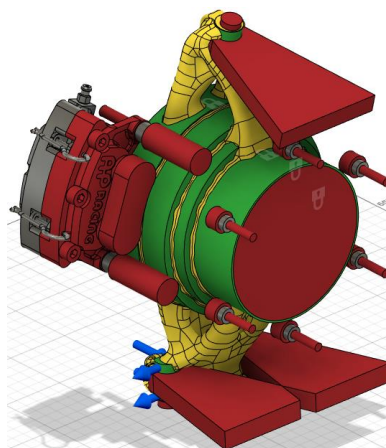


Figure 16: Fusion 360 starting geometry for producing second pass generative upright

3.3. Boundary Conditions

The fixed boundary conditions that were used for the generation are located at the bearing faces (see Figure 17). This is because the axle pivots about the two bearing faces when loading is applied at the contact patch of the wheel, so the moments generated are applied to these faces.

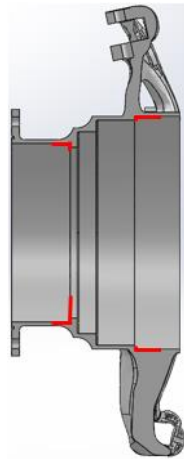


Figure 17: Upright cross-section with highlighted bearing faces

The upright was also validated within SolidWorks FEA, where different boundary conditions were utilised (see Figure 18). These boundary conditions were even more reflective of the real-world loading conditions, where the upright is constrained by the wishbone mounts (yellow). This approach was not applicable in generative design, as the gearbox and wheel hub assembly (green) could not successfully be integrated into the generation assembly without causing errors. A solid approximation of the gearbox bearings (blue) was utilised, as using the bearing CAD models themselves would consume unnecessary time and computing resources.

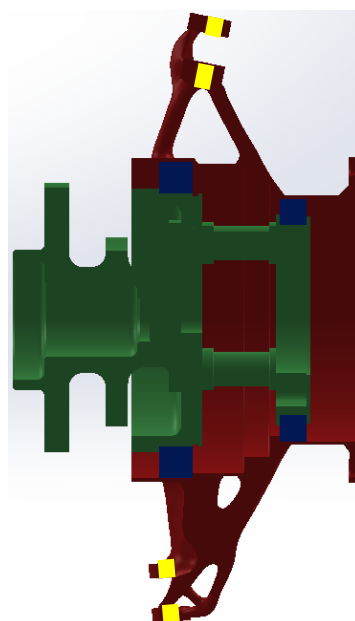


Figure 18: SolidWorks FEA validation geometry

3.4. Loading Conditions for Generation

The loading conditions were calculated as a moment of forces assumed to be acting on the contact patch centroid of the tyre. A 2g lateral load, 3g unexpected bump load, and 2g deceleration load were considered simultaneously, as well as peak gearbox torque, peak braking torque, and peak motor torque. A 3g inertial motor bump acceleration load was also applied, considering the mass of the motor.

The mass transfer during cornering was calculated as follows:

$$\text{Proportion of load transfer} = \frac{g_c \times H_{COG}}{W_t}$$

$$\frac{2 \times 0.27364}{1.2822} = 0.426829$$

$$\text{Proportion of load on inner wheels} = 0.5 - 0.426829 = 0.0732 \text{ (3 s.f.)}$$

$$\text{Proportion of load on outer wheels} = 0.5 + 0.426829 = 0.927$$

So, considering a full car mass of 350kg including the driver:

$$m_{s,i} = \frac{350 \times 0.0732}{2} = 12.8 \text{ kg per inside wheel}$$

$$m_{s,o} = \frac{350 \times 0.927}{2} = 162 \text{ kg per outside wheel}$$

The figure of 350kg was decided upon to reflect the vehicle mass plus average driver mass, with a margin to allow for additional driver mass.

So, considering a 2g lateral and 3g vertical load:

$$F_{l,i} = g_c \times 9.81 \times 12.8 = 251 \text{ N}$$

$$F_{v,i} = g_v \times 9.81 \times 12.8 = 377 \text{ N}$$

$$F_{l,o} = g_c \times 9.81 \times 162 = 3180 \text{ N}$$

$$F_{v,o} = g_v \times 9.81 \times 162 = 4770 \text{ N}$$

A 2g lateral acceleration load reflects the maximum observed cornering g-force of the vehicle in practical testing. The 3g Bump load reflects an exaggerated bump force load in the event of hitting a large imperfection on the track surface, or in the event that the vehicle leaves and re-joins the track surface. These loads are applied from the contact patch of the tyre, and moments were calculated about the rotational mid-point of the gearbox in the relevant plane. The mid-point between the two gearbox bearings was considered to be the point of rotation (see Figure 19).

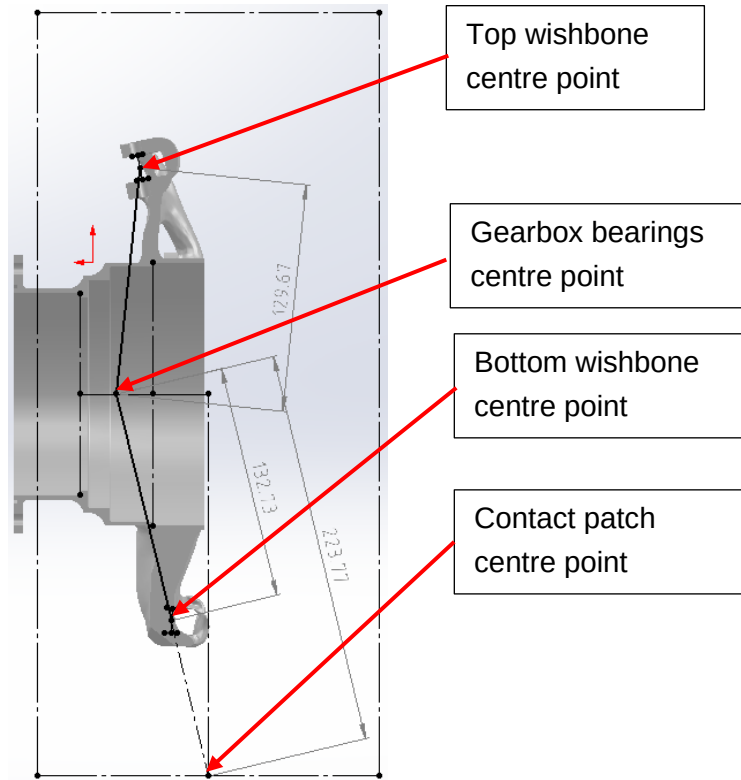


Figure 19: Dimensional arrangement for calculating wishbone reaction forces to lateral and vertical contact patch loads

The 2g deceleration load was calculated as follows:

$$F_d = 2 \times 9.81 \times 350 = 6867 \text{ N}$$

This load reflects forces seen if an object is impacted, such as a cone, on the track. Practical data is unavailable on this, so an exaggerated approximation is employed. This load was applied to the wishbone mounting points as a reaction force, assuming the impact is applied at the central side of the tyre and the gearbox acts as a beam, with the wishbone mounting points acting as pinned beam boundary conditions.

Braking reaction force on the caliper mount were calculated as follows:

$$F_b = \frac{T_b}{r_r} = \frac{1000}{0.114} = 8770 \text{ N}$$

Braking torque was approximated as being 1000 Nm, which is a slightly exaggerated figure for the purpose of design safety. The force is calculated at the distance the caliper mounts lie from the central rotational axis of the wheel hub and applied across the two caliper mounting bosses.

Peak gearbox torque was calculated as follows:

$$T_p = T_{p,m} \times \text{gearbox reduction ratio}$$

$$T_p = 30 \times 14 = 420 \text{ Nm}$$

This force was calculated as a reaction force by the ring gear exerted on the gearbox casing and was applied as a reaction force on the top and bottom wishbone mounting points about the central bearing axis of the gearbox.

Motor torque reaction force on the motor mounts was calculated as follows:

$$T_{m,peak} = 30 \text{ Nm}$$

$$F_{m,torque} = \frac{30}{0.0715} = 419.6 \text{ N}$$

The maximum motor output torque was obtained from the manufacturer's specification sheet, and the force was calculated by considering the distance between the central rotational axis of the motor and the centre of the motor mounting fastener holes (see Figure 22). It was applied as a distributed force across all of the motor mounting bosses.

3g Inertial motor acceleration force was calculated as follows:

$$m_m = 4.295 \text{ kg}$$

$$F_{m,inertial} = m_m \times 3 \times 9.81 = 126.4 \text{ N}$$

$$\text{moment about the central bearing axis} = 126.4 \times 0.12156 = 15.37 \text{ Nm}$$

$$\text{reaction force from motor bosses to counter this moment} = \frac{15.37}{0.12156 - 0.06275} = 261 \text{ N}$$

The inertial motor mount acceleration force simulated the force created by the inertial mass of the motor experiencing the suspension bump force mentioned previously. It was calculated as a moment about the central bearing axis of the gearbox, with the force being applied from the motor centre of mass (see Figure 20) and applied as a vector reaction force to each of the motor mounting bosses.

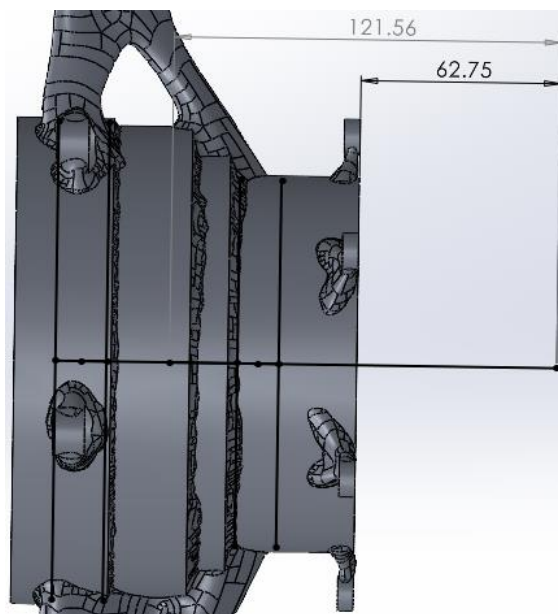


Figure 20: Dimensional arrangement for calculating motor inertia reaction forces

The outer wheel loading figures were used for generative model creation, since these are the worst-case loads that the upright will experience. The wishbone reaction forces were broken down as an axial force and a normal force (see Figure 21). The normal force is neglected on the bottom wishbone as there is no mechanical resistance to normal displacement because the car utilises pull-rod suspension.

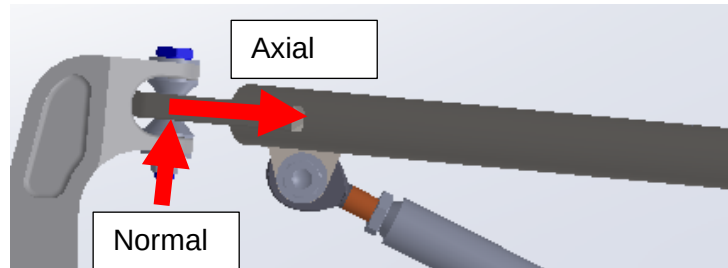


Figure 21: Illustration of axial and normal forces applied to a wishbone [3]

After evaluating the forces applied at the contact patch, the following wishbone reaction forces were determined using simple trigonometry (see Appendix 1):

Total top wishbone axial reaction force: 8344 N

Total top wishbone normal reaction force: 5824 N

Total bottom wishbone axial reaction force: 8287 N

Maximum gearbox forces were calculated by evaluating the gearbox moment about the central axis of the gearbox (see Figure 22) and then applying it at the wishbones as a reaction force.

$$\text{Top wishbone peak gearbox torque reaction force} = \frac{420}{0.12875} = 3262 \text{ N}$$

$$\text{Bottom wishbone peak gearbox torque reaction force} = \frac{420}{0.13362} = 3143 \text{ N}$$

Maximum frontal impact forces were calculated by treating the gearbox as a beam, with the load being applied centrally to the main body, and the consequential reaction forces being calculated at each of the wishbone attachment points (see Figure 22).

Top wishbone impact reaction force = 3433 N

Bottom wishbone impact reaction force = 3433 N

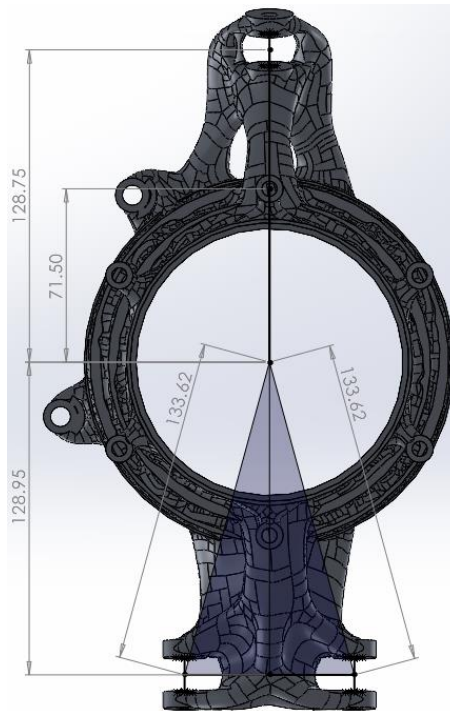


Figure 22: Dimensional arrangement for calculating wishbone reaction forces to peak gearbox torque and motor boss reaction forces to peak motor torque

3.5. Validation

Whilst Fusion 360 utilises its own FEA program to carry out the generative design process, which ensures that the factor of safety is above two. The upright was then validated separately in SolidWorks FEA. This was to provide reassurance of the results through analysis in a different program which is capable of utilising ISO clipping to identify areas of high stress and eliminate the inclusion of singularities. By utilising this secondary software, a customised and refined mesh was able to be created, which naturally yielded more accurate FEA results.

Areas of minimum stress were also able to be identified in the software, which allowed for a comparison of material volume utilization to be established between the generatively designed uprights and the previous design.

Maximum displacement was also determined. This allowed comparison between the designs, as well as the previous upright. Maximum displacement is an important factor for the design, as many faces within the gearbox have tight tolerances, so large displacements in these locations could cause unnecessary friction, gear tooth binding, and eventual failure.

4. Results, Data Analysis, and Discussion

4.1. Material Testing

Tensile material testing was carried out on nine AlSi10Mg testing samples. Five of the samples were left untreated, while four of the samples were heat treated by annealing at 263 degrees Celsius for 10 minutes. From published literature, it was decided that this temperature would be appropriate as this would achieve increased elongation whilst not disrupting the silicon matrix within the material, which would reduce yield strength drastically.

Material testing was successfully performed, and the yield strength and elastic modulus determined for each sample (see Table 1). However, it was discovered that the values obtained differed greatly from the expected values. Other published literature suggests that the yield strength of AlSi10Mg should be in the region of 250 MPa for the untreated samples, which is 54.02% greater than the value obtained here. Yield strength should also be approximately 225 MPa for the treated samples, which is 49.62% greater than the value obtained here. Furthermore, the observed young's modulus is approximately 290% greater than the expected value. The 0.2% offset method was utilised to calculate the yield strength of the material, while the slope of the elastic region was used to calculate the elastic modulus (see Appendix 5).

This discrepancy could be due to a number of reasons, but it was speculated that the main cause was porosity and inclusions within the sample material. Significant voids at the fracture site could be observed on many of the samples (see Figure 23). This would explain the unusually brittle behaviour of the material at an unexpectedly low yield strength, as this would cause premature fracture propagation. It would be beneficial to modify the printer settings and repeat the testing until a more typical value is obtained. Thus, this testing data has not been utilised in this report to produce the generative designs, as it is not representative of the true material properties. Instead, more accurate figures from published literature have been used [25].



Figure 23: Picture showing inclusions and porosity in the tensile testing samples

Sample number (untreated)	Yield strength (MPa)	Elastic modulus (GPa)
1	126.12	306.63
2	127.19	309.63
3	101.81	248.67
4	119.32	287.89
5	100.24	241.97
Average	114.94	278.99
Sample number (annealed)		
1	105.44	252.79
2	130.94	315.49
3	102.46	237.44
4	114.56	277.40
Average	113.35	270.77

Table 1: AlSi10Mg material testing data

4.2. Final Designs

Final designs have been successfully produced in both aluminium (AlSi10Mg) and titanium (Ti6Al4V) material variants, and are pictured below (see Figure 24 and Figure 25) in context with the other components necessary to complete a corner power unit. Integrated in both designs are the necessary wishbone attachment points, brake caliper attachment points, motor attachment points, and internal features to accommodate the gearbox components.

Extra material has been added to all critical faces in order to allow for final machining. This is necessary because the AM process will leave an imprecise and rough surface finish on all exposed surfaces due to semi-melted powder inclusion at the edge of the weld pool. Critical faces are required externally for accurately mounting the wishbones, brake caliper and motor. More importantly, surfaces with close tolerances and a high degree of concentricity are required for mounting the internal gearbox bearings, and the ring gear for the compound planetary geartrain. Smooth running and proper alignment of the gearbox is imperative to efficient and reliable power output to the wheels.

Wheel alignment must also be considered when designing an upright suspension member. Castor must be set to a desired value, and camber and toe must be fully adjustable in order to allow proper tuning of the vehicle dynamics. The castor angle is set by the wishbone geometry tying the upright to the chassis. The designs featured in this report mirror the previously implemented castor angle, to ensure that it is within the desired range. Adopting the same attachment point locations as before also allows for easy implementation into the car. Toe adjustment is achieved within the wishbone geometry by varying the length of the tube attached to the rearmost lower wishbone attachment point. Camber adjustment is achieved by utilising eccentric bolts on the chassis side of the lower wishbone. This is a technique successfully utilised on many production vehicles and serves as a robust, low cost, and easily adjustable solution.

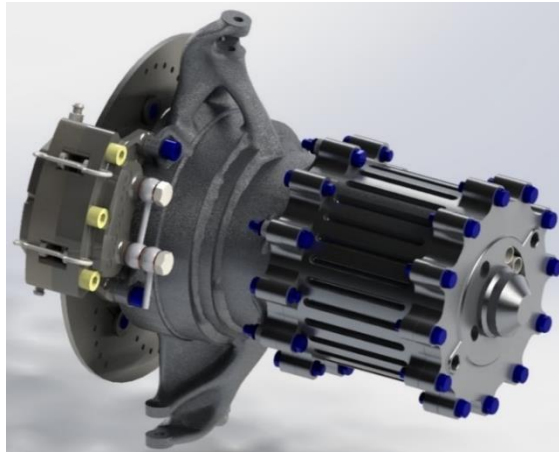


Figure 24: Final AlSi10Mg upright design in context

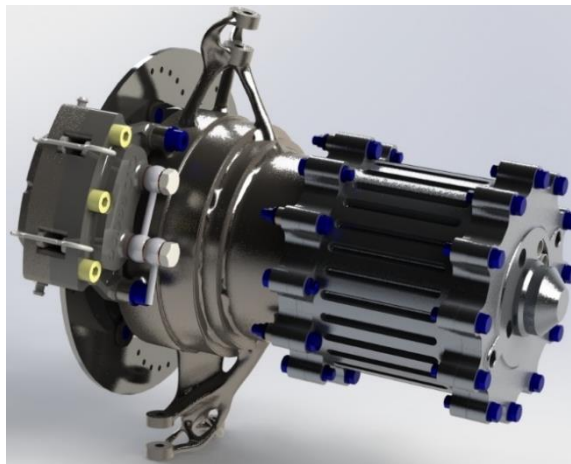


Figure 25: Final Ti6Al4V upright design in context

4.3. Mass Analysis

The mass reduction profile over the 55 generative design software iterations (see Figure 26) clearly demonstrated the effect of each iteration on the mass of each upright. Furthermore, the positive impact of feeding the first pass design back into the software is identifiably apparent.

The aluminium upright experienced an 84.8% (3.1kg) reduction in mass from iteration 0 to iteration 55. 93.7% of the mass reduction was achieved in the first pass (iterations 0 to 30), and 6.3% of the mass reduction was achieved in the second pass (iterations 35 to 55). Meanwhile, the FOS was reduced by 53.7% from 4.318 to 2. 71.6% of this FOS reduction was achieved in the first pass, and 28.4% was achieved in the second pass.

The titanium upright experienced an 92.1% (4.9kg) reduction in mass from iteration 0 to iteration 55. 86.2% of the mass reduction was achieved in the first pass, and 13.8% of the mass reduction was achieved in the second pass. Meanwhile, the FOS was reduced by 87.4% from 15.943 to 2. 43.1% of this FOS reduction was achieved in the first pass, and 56.9% was achieved in the second pass.

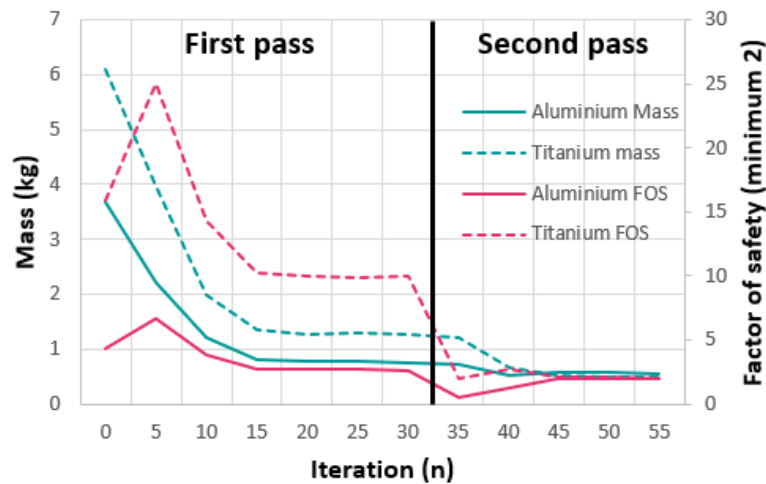


Figure 26: A graph demonstrating the mass reducing and factor of safety converging as generative iterations are carried out

When comparing the mass of each designed upright to the previous multi-piece billet upright, clear differences can be identified (see Table 2). This mass was evaluated using SolidWorks as, upon comparison with Fusion 360, it produced a more accurate result (see Appendix 3).

The generatively designed and additively manufactured titanium upright represents a 69.1% (1.522kg) mass reduction. As such, in an AWD configuration, this would reduce total vehicle mass by 6.088kg (2.44% of total vehicle mass). While such a small reduction in total vehicle mass may seem insignificant, a greater gain can be seen in the reduction of unsprung mass at 7.96%, which will have a significant positive impact on vehicle dynamics (see Table 2).

The generatively designed and additively manufactured aluminium upright represents a 69.9% (1.54kg) reduction in mass. In an AWD configuration, this would reduce total vehicle mass by 6.164kg (2.47% of total vehicle mass). Much like the titanium upright design, a significant reduction in unsprung mass can also be observed, but to an even greater extent at 8.06%, which will likewise have a significant positive impact on vehicle dynamics (see Table 2).

Upright design	Mass (kg)	Total unsprung mass per corner (kg)
2023 multi-piece billet upright	2.204	19.114
AM titanium Ti6Al4V	0.682	17.592
AM aluminium AlSi10Mg	0.663	17.573

Table 2: Comparison table of upright design mass

While both upright designs satisfy the success criteria of this report, by being at least 20% lighter than the previous design, it is clear that the aluminium upright outperforms titanium in terms of absolute mass. This is due to the fact that there is a significant amount of fixed material volume in the cylindrical gearbox housing. While this could theoretically be made thinner in the titanium design, it would greatly compromise the practicality and manufacturability of the design. While a gearbox casing thinner than 1.5mm is technically acceptable under the loading conditions specified within this report, the fragility of a thinner casing than this cannot be underestimated. This is especially the case, if by chance an

inclusion or void is introduced in the printing process because this would produce a significant weak point in the gearbox casing.

4.4. Stress Analysis

Both newly designed uprights were successfully validated in SolidWorks FEA (see Appendix 3) and, after singularities were discounted from the analysis, both displayed a FOS of two or greater under a worst-case loading scenario.

The aluminium upright experienced a maximum observed stress of 125 MPa, which is 20% less than the maximum stress experienced by the multi-piece billet upright. However, the maximum stress region observed on the multi-piece upright is located on the top titanium part, and as such has a FOS of around 5.3. Although this is positive in terms of part strength, it is a negative indication in terms of mass optimisation. The minimum observed stresses also support this, as the multi-piece upright experiences 99.7% less minimum stress than the aluminium upright. This suggests that the multi-piece upright poorly utilises its material volume, as a significant amount of the material present is under-stressed.

The titanium upright experienced a maximum observed stress of 475 MPa, which is 303% greater than the maximum stress experienced by the multi-piece billet upright. It also experiences 1362 times greater minimum stress than the multi-piece upright. This suggests, much in the same way as the aluminium design, that the titanium design utilises its material volume more effectively than the multi-piece upright. Furthermore, it also experiences 395% greater minimum stress than the aluminium design, which could suggest that the material volume is utilised even more effectively in the titanium design.

Both designs satisfy the safety success criteria by having an FOS greater than or equal to two meaning either design is suitable for use, according to the success criteria specified.

Upright design	Max. observed stress (MPa)	Min. observed stress (MPa)
2023 multi-piece billet upright	157	0.000058
AM titanium Ti6Al4v	475	0.079
AM aluminium AlSi10Mg	125	0.02

Table 3: Comparison table of upright design maximum and minimum observed stress

4.5. Displacement Analysis

Displacement analysis was also performed in SolidWorks FEA. This allowed the determination of the magnitude and location of areas of maximum displacement. If areas of large displacement were present on critical faces within the gearbox, it could cause undesirable compression of the bearings or ring gear, or even interfere with the rotating assembly.

The aluminium upright experienced a maximum observed displacement of 0.27mm, which is 450% greater than the maximum displacement experienced by the previous upright. This maximum displacement occurs on the motor mounting bosses (see Appendix 3). This could then cause issues with sun gear alignment inside the gearbox, as it relies on alignment pins within three of these mounting bosses for proper concentricity alignment. This indicates that

further manually added reinforcement above 2 FOS is likely to be required on these areas, in order to reduce the material displacement under load.

The titanium upright experienced a maximum observed displacement of 0.9mm, which is 1500% greater than the previous upright. This maximum displacement also occurs on the motor mounting bosses (see Appendix 3). The aforementioned issues with regard to the aluminium upright could also be caused in the case of the titanium upright. Benefits would also be felt from further manually added reinforcement on these mounting points, as high deflection in this area is undesirable, even if the factor of safety criteria is achieved. However, as the maximum displacement is greater, and the density of titanium is greater, this would further increase the mass disparity between each of the designs.

Upright design	Maximum observed displacement (mm)
2023 multi-piece billet upright	0.06
AM titanium Ti6Al4v	0.9
AM aluminium AlSi10Mg	0.27

Table 4: Comparison table of upright design maximum observed displacement

4.6. Material Evaluation

AlSi10Mg is considered in the industry to be a mid-strength ‘workhorse’ alloy, with the powder cost being relatively inexpensive. Relatively few supports are required to tie the part to the bed of the printer, and are not required in preventing part warping during manufacture. The material is then able to be annealed, which reduces strength slightly, but increases elongation at failure. As such, the upright is made less susceptible to fatigue failures, and less likely to exhibit catastrophic brittle failure under use.

Whilst Ti6Al4V has a significantly higher yield stress than AlSi10Mg, it presents several manufacturing challenges. Firstly, the powder is 257% more expensive than AlSi10Mg. This may not seem especially influential when considering the part alone, but the entire volume of the printer must also be filled with powder to surround the part, meaning that a very significant initial investment has to be made to purchase a large volume of powder. Secondly, a titanium part will tend to curl due to stresses within the material created during printing [26]. As such, robust and plentiful supports are required for every freely suspended piece of the part. However, due to the robust nature of the material, support removal from the bed of the printer, and the part itself, requires wire electrical discharge manufacturing (EDM). While it is possible to snap the supports from the part, this is likely to cause fractures in the material, which will eventually lead to fatigue cracking in the long term. This material also benefits from stress relieving, which reduces the internal stresses in the part created during the printing process. This is ideally carried out whilst the part is still attached to the print bed by the support structures, as it may warp if carried out after being separated. After being separated from the print bed, the part would benefit from annealing at 850 degrees Celsius, and can then be furnace cooled, air cooled, or water quenched, depending on the desired yield strength and elongation [18].

As the part is applied in a motorsport application, higher elongation at the cost of a slight strength decrease is favourable, as a deformed part due to unexpected loads is safer than a part completely failing unexpectedly due to brittle fracture.

4.7. Manufacturing Technique Evaluation

SLM and DED technologies were evaluated against each-other as appropriate manufacturing methods. SLM allows for a higher resolution by utilising smaller layer thickness, but this incurs a significant time cost. DED is substantially faster, as material deposition and melting occurs simultaneously. This means that it is generally more appropriate for larger structures. However, SLM is more appropriate for complex structures as much of the part geometry is supported by the auxiliary powder surrounding the part, as well as having a smaller melting pool. DED requires a great deal more support, which causes issues for post-processing, especially when the part geometry is complex. Consequently, SLM has been evaluated as the most appropriate technology.

4.8. Material Selection

While AlSi10Mg exhibits favourable mass properties, demonstrated by the findings in this report, other alloys such as Scalmolly could also be utilised. This material has a very similar density, whilst being just over twice as strong, and is also Fédération Internationale de l'Automobile (FIA) approved. Thus, part mass could be decreased even further. However, this material is approximately 23 times more expensive than AlSi10Mg, which could demonstrate a significant practical expense issue. A relatively low temperature heat treatment can also be applied in order to reduce the likelihood of catastrophic brittle failure due to imperfections or fatigue.

AlSi10Mg is also significantly easier to manufacture when compared to Ti6Al4V. Supports are not required throughout the entire part, as they are only required to attach the part to the print bed. For titanium, the internal stresses of the part during the printing process are high. This means robust and plentiful supports are required to prevent warping. These supports present significant time and monetary expenses due to the process of removal after manufacture.

The aluminium design also features a lower maximum displacement. This is desirable as the motor and gearbox assembly relies upon a high level of geometric alignment. If deformation misaligns the motor, then the gear assembly is more likely to experience interference, which could result in increased wear or even failure. As previously addressed, additional reinforcement above the factor of safety is likely to be required on the motor mounts, but less mass penalty will be incurred for this compared to the same level of reinforcement on the titanium design.

Because of this, the AlSi10Mg generated design is the more favourable of the two, in terms of the four pertinent criteria: mass, displacement, cost, and manufacturability.

4.9. Manufacturing

Manufacturing quotations have been obtained from two third party additive manufacturing companies, Xometry and Protolabs. The quotes were £1,621.41 and £2,988.54 per part respectively. Discussions have also been held with an internal faculty additive manufacturing group. However, it has been concluded that, due to the complexity and size of this part, UoNRT would be better served by a third-party facility. This is because they have access to more modern and reliable machinery, as well as being already equipped with the powder volume necessary to complete this project. A third-party would also absorb the cost of any machine failures during the printing process. Furthermore, it is likely that a suitable sponsor could be found to reduce the financial impact of this manufacturing, as demonstrated by OBR collaborating with the national centre for additive manufacturing (NCAM) to produce their AM upright. It is also worth noting that this design could be produced on a 5-axis CNC machine utilising 7000 series billet aluminium. This would be more expensive than additive manufacturing, but the final part would be undeniably stronger.

An engineering drawing was also produced (see Appendix 6) detailing all dimensions for post-processing machining required to appropriately interface with the existing internal gearbox, brake, and motor systems (completable in-house by the machining cell in the L2 workshop or the precision manufacturing centre, or by Xometry or Protolabs on request).

5. Conclusions

Two designs have been successfully created in titanium Ti6Al4V and aluminium AlSi10Mg as a part of this report. Many generative design iterations were carried out within Fusion 360 for each material case, and the best iteration for each material was selected by evaluating FOS and mass properties. Both designs achieve the success criteria established in the aims and objectives of the report, as they integrate the upright and gearbox casing into one cohesive component. An FOS 2 or greater under the worst-case combination of load cases is achieved in both cases, and both new designs are at least 20% lighter than the previous design.

While practical material testing was carried out, it did not yield tangible results. This meant more accurate figures from a third-party were used within the generative design software instead. A valid heat-treatment process was also investigated which yielded a slight decrease in yield stress and elastic modulus, indicating that the samples became more ductile.

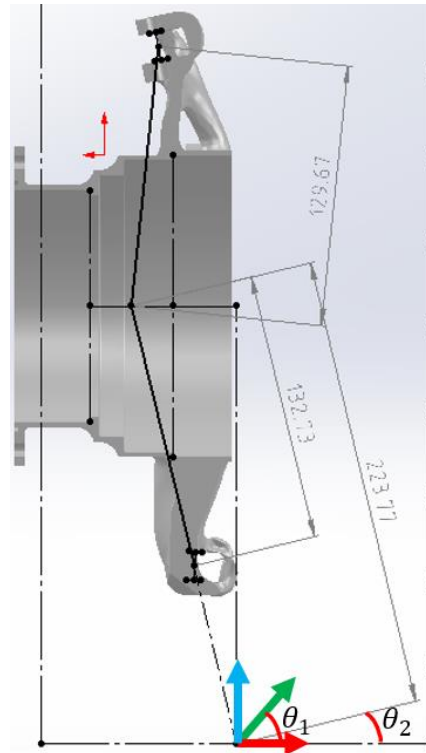
Two possible third-party manufacturers were established with quoted prices, giving next year's race team a clear pathway to implement the designs created in this report. Based on the evidence presented by this study, there is clear reasoning that a significant reduction of mass is achievable in the upright assembly by adopting an aluminium additively manufactured component in place of the current inefficient assembly of multiple machined components. Overall, this will improve vehicle dynamics, resulting in a more competitive car, yielding more favourable results in the dynamic testing categories of the FS competition. Hopefully this, in tandem with other advancements, will continue the upward trajectory of the team in coming years.

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Appendix 1 – Wishbone Reaction Force Calculations



Calculating the moment and axial force applied by the contact patch to the central axis of the gearbox:

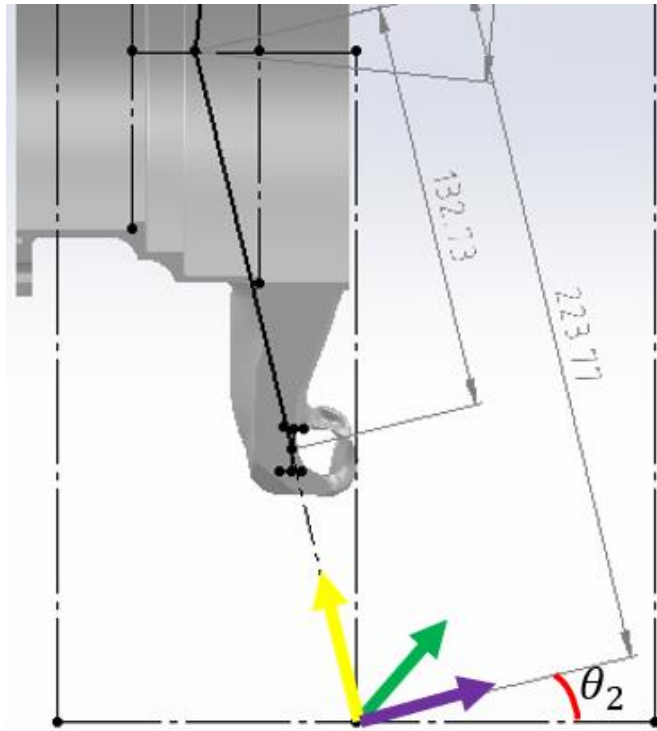
$$F_l(\text{red arrow}) = 3180 \text{ N}$$

$$F_v(\text{blue arrow}) = 4770 \text{ N}$$

Calculating the combined vector force:

$$\theta_1 = \tan^{-1} \frac{4770}{3180} = 56.3^\circ$$

$$\text{Combined vector force (green arrow)} = \sqrt{F_l^2 + F_v^2} = 5733 \text{ N}$$



Calculating the moment inducing force and axial force:

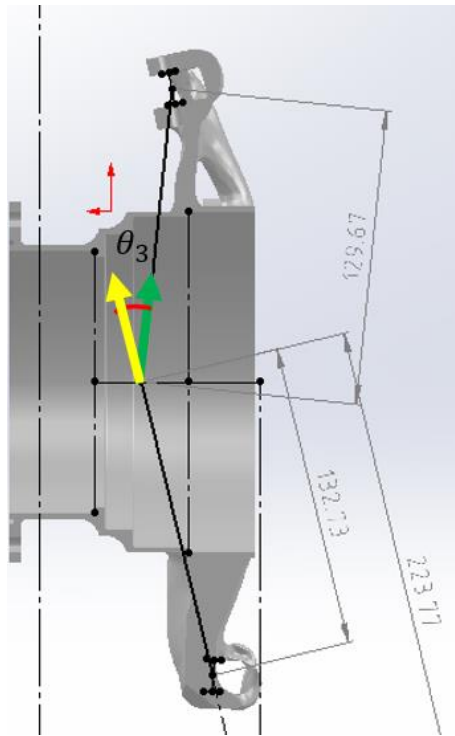
$$\theta_2 = 13.51^\circ \text{ from measurement}$$

$$F_a \text{ (yellow arrow)} = 5733 \sin(\theta_1 - \theta_2) = 3894 \text{ N}$$

$$F_{\text{moment}} \text{ (purple arrow)} = 5733 \cos(\theta_1 - \theta_2) = 4207 \text{ N}$$

So, the moment acting about the centre axis of the gearbox:

$$T = 4207 \times 0.23377 = 983.47 \text{ Nm}$$

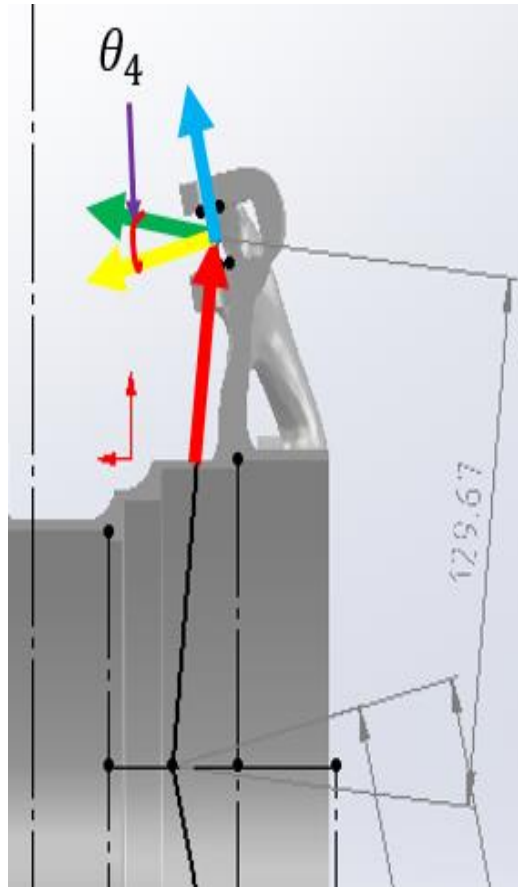


Calculating the axial force transferred to the top wishbone pickup point:

$$\theta_3 = 19.61^\circ \text{ from measurement}$$

$$F_a \text{ (yellow arrow)} = 3894 \text{ N}$$

$$F \text{ (green arrow)} = 3894 \cos(19.61) = 3668 \text{ N}$$



Moment induced force at pickup point (green arrow)

$$F_{moment\ induced} = \frac{983.47}{0.12967} = 7584\ N$$

$$\theta_4 = 17.92^\circ$$

Resolved into axial wishbone (yellow arrow) and normal wishbone (blue arrow) directions:

$$F_{a,from\ moment} = 7584 \cos(17.92) = 7216\ N$$

$$F_{n,from\ moment} = 7584 \sin(17.92) = 2334\ N$$

Axially transferred force at pickup point (red arrow)

$$F_a = 3668\ N$$

Resolved into axial wishbone (yellow arrow) and normal wishbone (blue arrow) directions:

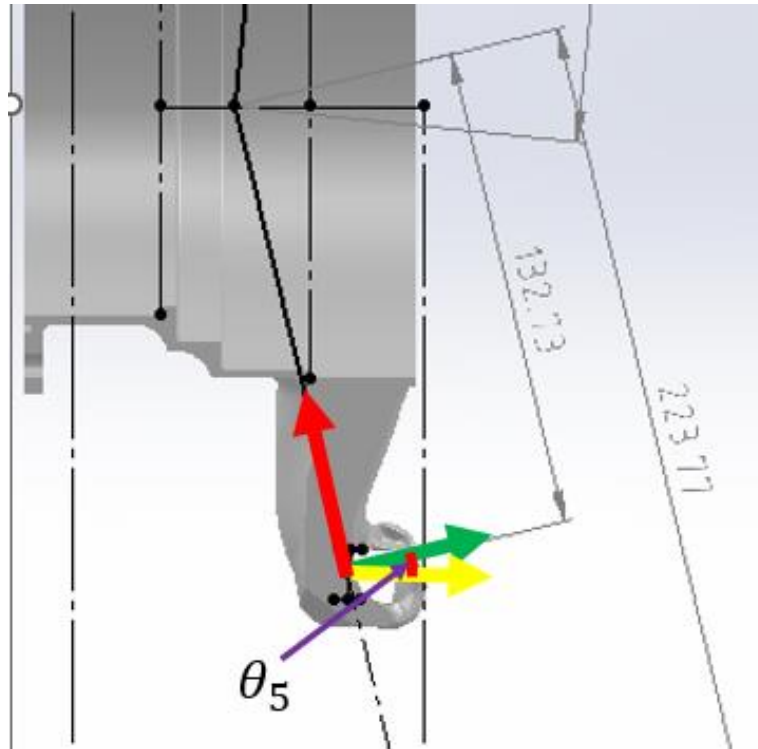
$$F_{a,from\ axial\ force} = 3668 \sin(17.92) = 1128\ N$$

$$F_{n,from\ axial\ force} = 3668 \cos(17.92) = 3490\ N$$

So, total axial wishbone (yellow arrow) and normal wishbone (blue arrow) forces:

$$F_{a,top\ wishbone} = 7216 + 1128 = 8344\ N$$

$$F_{n,top\ wishbone} = 2334 + 3490 = 5824\ N$$



Moment induced force at bottom wishbone pickup point (green arrow):

$$F_{moment\ induced} = \frac{983.47}{0.13273} = 7410\ N$$

$$\theta_5 = 19.61$$

Resolved into axial wishbone (yellow arrow) direction:

$$F_{a,from\ moment} = 7410 \cos(19.61) = 6980\ N$$

Axially transferred force at pickup point (red arrow)

$$F_a = 3894\ N$$

Resolved into axial wishbone (yellow arrow) direction:

$$F_{a,from\ axial\ force} = 3894 \sin(19.61) = 1307\ N$$

So, total axial wishbone forces (yellow arrow):

$$F_{a,bottom\ wishbone} = 6980 + 1307 = 8287\ N$$

Appendix 2 – Table of Generative Design Software Iteration Results

	AlSi10Mg		Ti6Al4V	
Iteration	Mass	Min. Factor of Safety	Mass	Min. Factor of Safety
0	3.682	4.318	6.11	15.943
5	2.21	6.681	3.98	25.045
10	1.212	3.847	1.995	14.316
15	0.806	2.682	1.369	10.298
20	0.772	2.68	1.28	10.015
25	0.784	2.678	1.29	9.845
30	0.755	2.658	1.258	9.936
35	0.727	0.54	1.206	1.998
40	0.515	1.283	0.662	2.766
45	0.572	1.938	0.524	2.136
50	0.579	2.041	0.501	2.071
55	0.558	2	0.48	2.012

Appendix 3 – SolidWorks Stress and Displacement Plots, and Mass Analysis

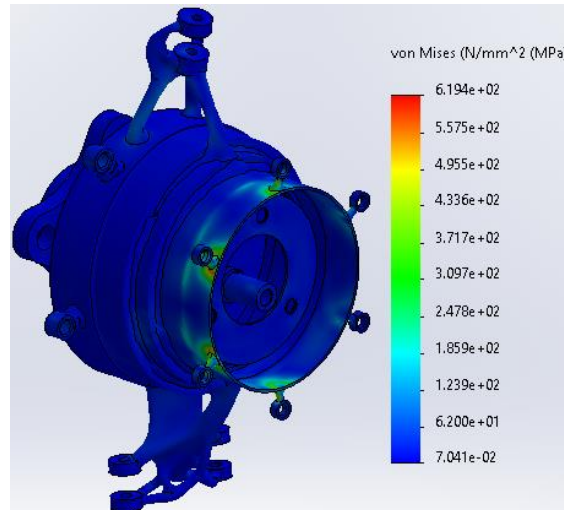


Figure 27: Ti6Al4V stress plot

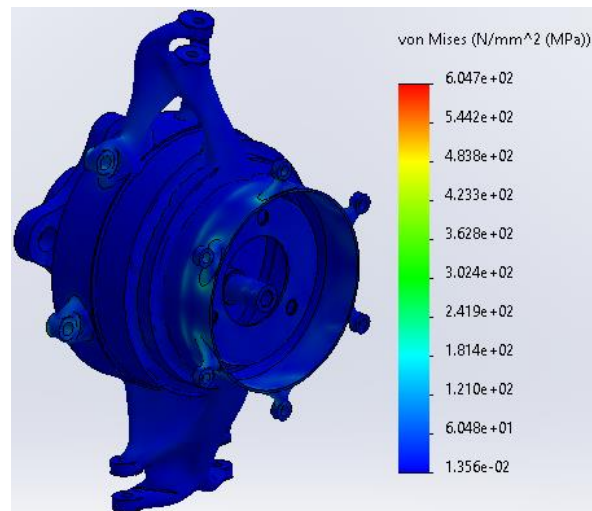


Figure 28: AISi10Mg stress plot

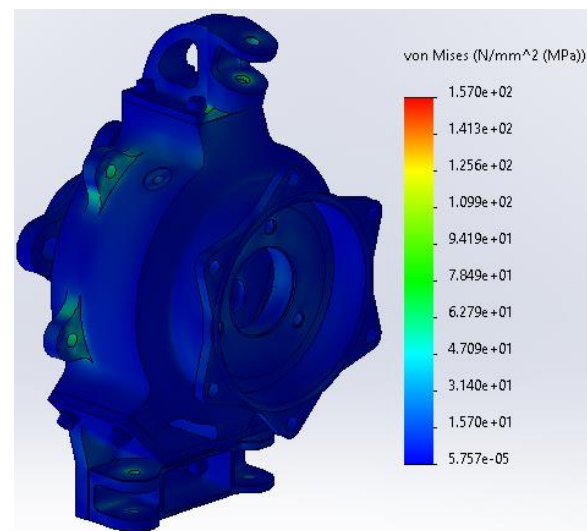


Figure 29: Previous upright stress plot

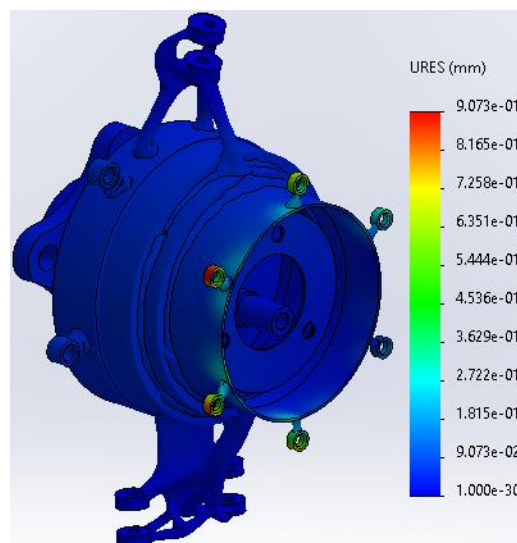


Figure 30: Ti6Al4V displacement plot

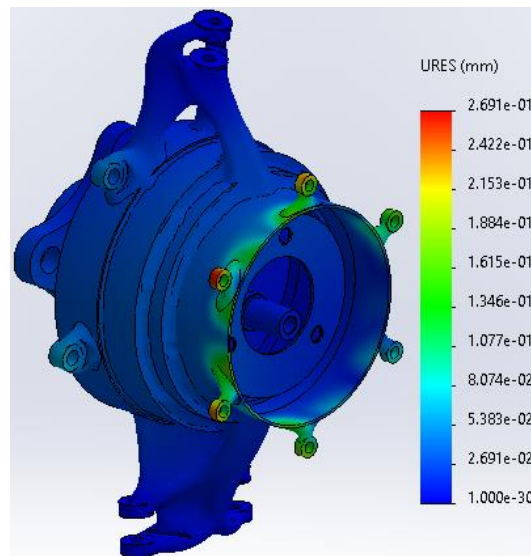


Figure 31: AISi10Mg displacement plot

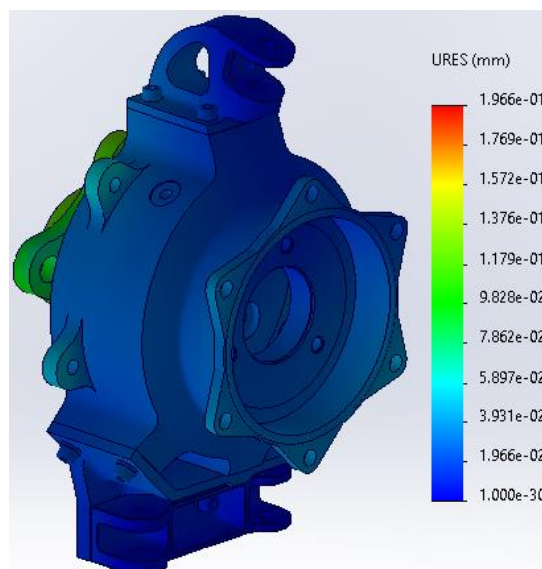


Figure 32: Previous upright displacement plot

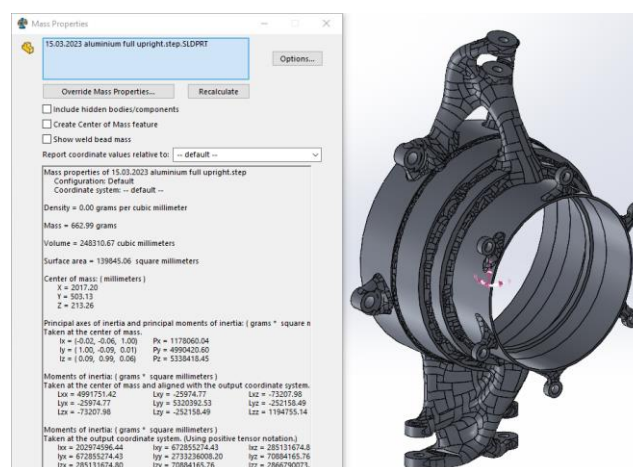


Figure 33: AISi10Mg upright mass analysis

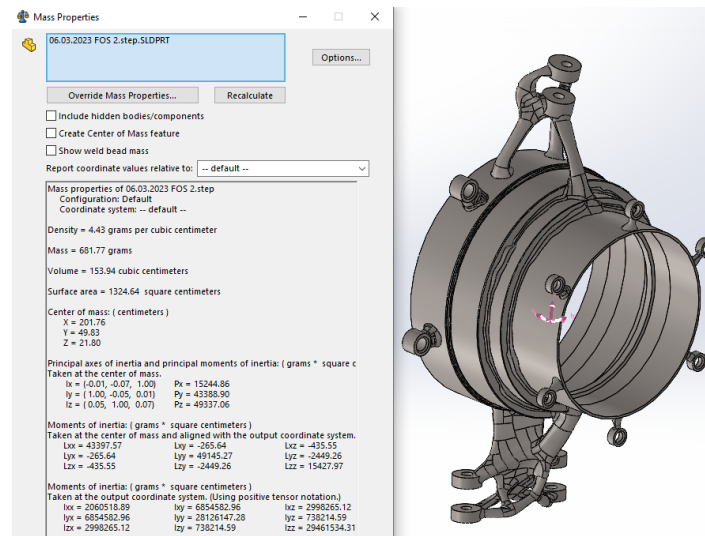


Figure 34: Ti6Al4V upright mass analysis

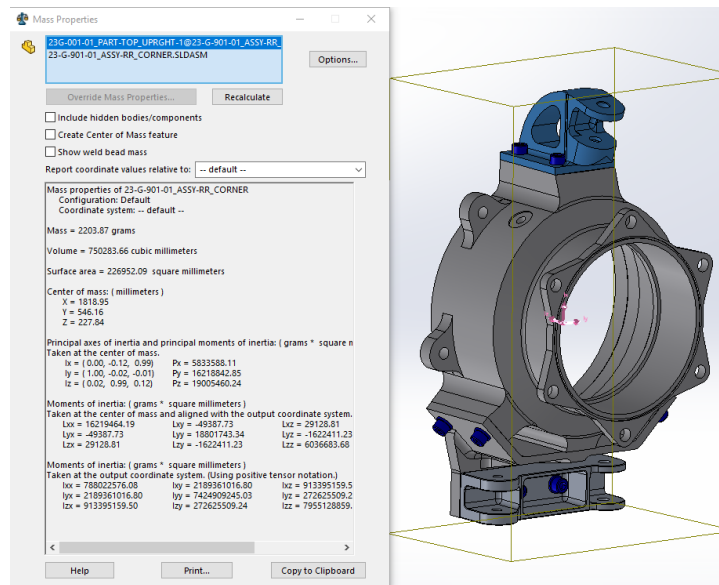
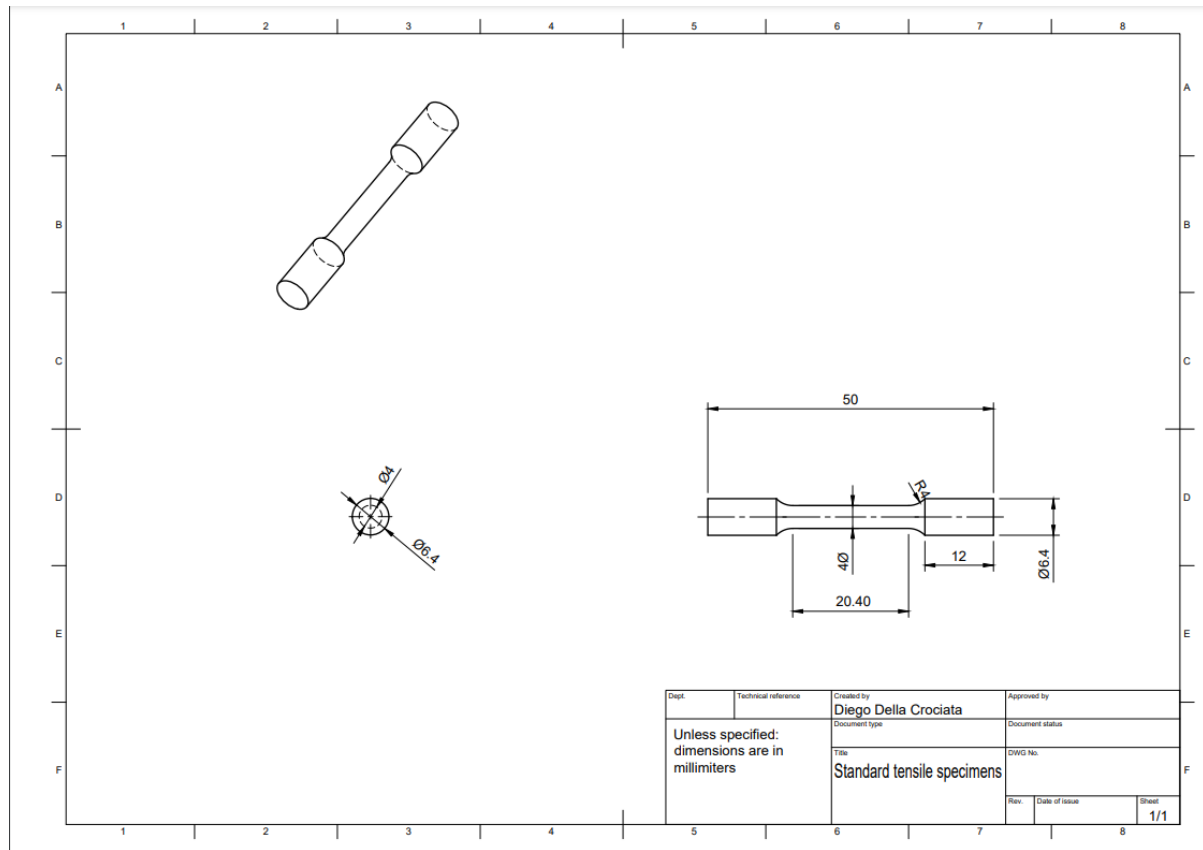
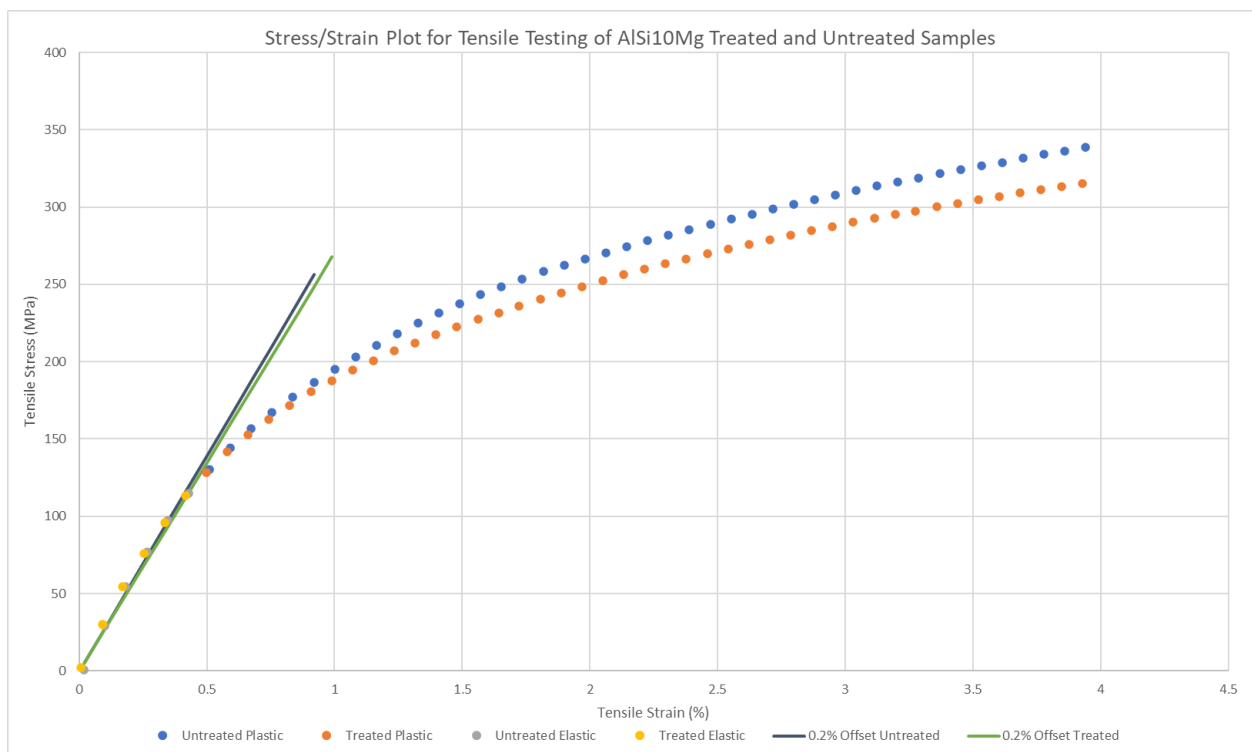


Figure 35: UoNRT 2023 multi-piece billet upright mass analysis

Appendix 4 – Tensile Test Sample Drawing

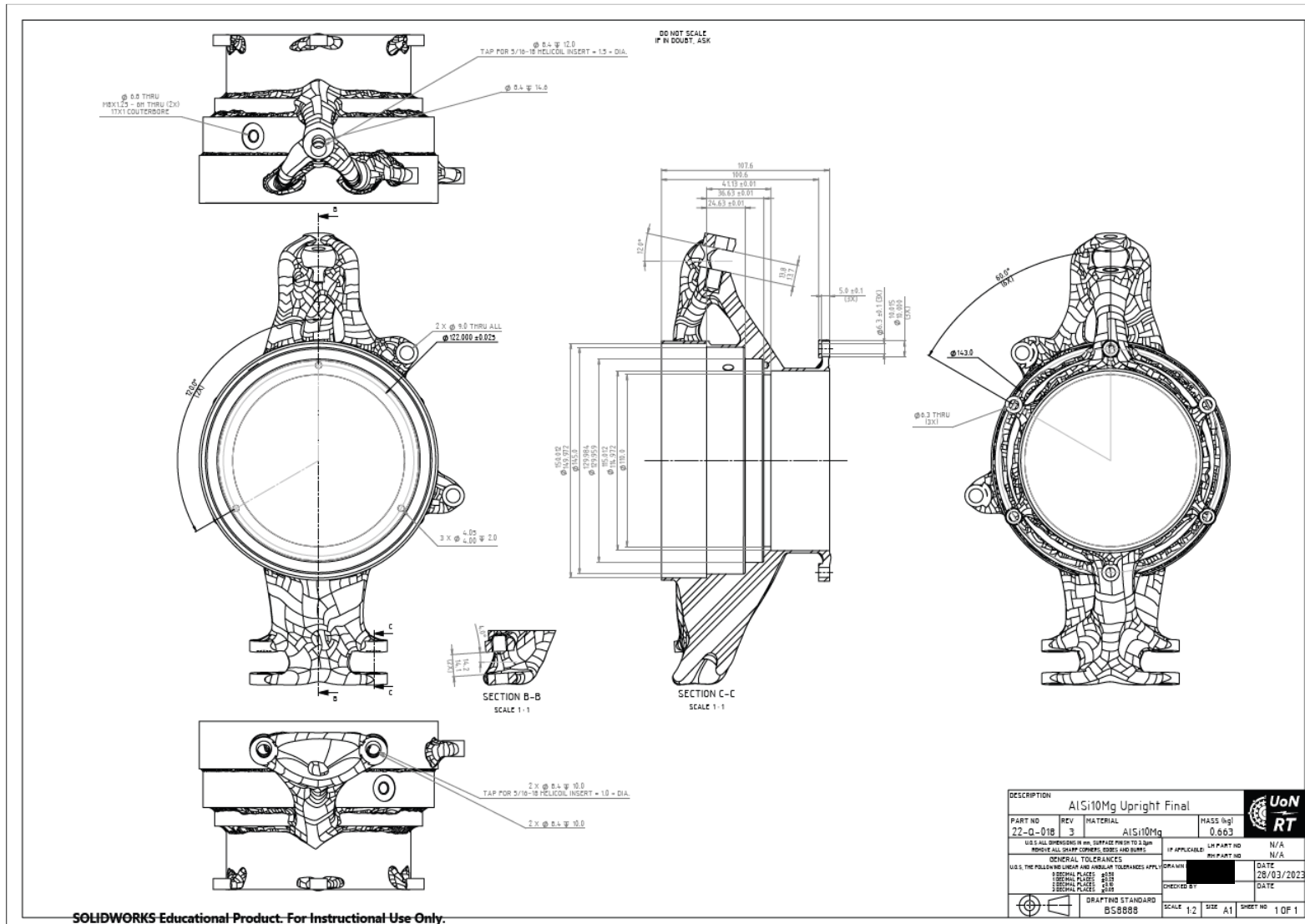


Appendix 5 – Stress/Strain Graph from AlSi10Mg Material Testing





Appendix 6 – Engineering Drawing for Post-Machining





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